

This overview outlines the [International Baccalaureate \(IB\) Diploma Programme \(DP\) Chemistry curriculum](#) offered at Chatsworth International School. Following the latest IB curriculum updates (first assessment 2025), the course focuses on a concept-based approach integrated through two organizing themes: **Structure** and **Reactivity**.

1. Curriculum Structure & Core Topics

The curriculum framework emphasizes how the structure of matter determines its chemical reactivity, which in turn transforms its structure. The core content is subdivided into three main areas for each theme:

Theme A: Structure (The nature of matter from simple to complex)

- **Models of the Particulate Nature of Matter:** Covers atomic models, electron configurations, and counting particles by mass (the mole concept) and ideal gases.
- **Models of Bonding and Structure:** Explores ionic, covalent, and metallic bonding, alongside molecular geometries and material structures.
- **Classification of Matter:** Focuses on periodic trends (the periodic table) and the classification of organic functional groups.

Theme B: Reactivity (How and why chemical reactions occur)

- **What Drives Chemical Reactions?:** Examines chemical energetics and thermodynamics, including enthalpy changes, energy cycles, and entropy.
- **How Much, How Fast, and How Far?:** Explores stoichiometry, chemical kinetics (reaction rates), and chemical equilibrium.
- **What Are the Mechanisms of Chemical Change?:** Investigates chemical pathways including proton transfer (acids and bases), electron transfer (redox processes), and electron sharing/pairing mechanisms.

2. Key Differences Between Standard Level (SL) and Higher Level (HL)

While both levels explore the foundational principles of Structure and Reactivity, they differ significantly in scope, intensity, and study hours:

- **Teaching Hours:** * **SL:** 150 total hours (110 hours of core syllabus content + 40 hours of experimental/practical programme).
 - **HL:** 240 total hours (180 hours of core and Additional Higher Level content + 60 hours of experimental/practical programme).
- **Depth and Breadth:** HL students delve into topics with greater mathematical and conceptual rigor. For instance, while SL covers basic energetics and kinetics, HL includes specialized subtopics such as *Entropy and Spontaneity*, advanced rate laws, and complex organic reaction mechanisms.

3. Assessment Components (First Assessment 2025 Onward)

The updated assessment model streamlines examinations by removing the previous Paper 3 (options) and integrating calculators and data booklets across all exam sections.

External Assessment (80% of final grade)

- **Paper 1 (36% Weighting):**
 - **Structure:** Divided into Paper 1A (Multiple-choice questions) and Paper 1B (Data-based and experimental analysis questions).
 - **Duration:** 1 hour 30 minutes for SL (55 marks) | 2 hours for HL (75 marks).
- **Paper 2 (44% Weighting):**
 - **Structure:** Consists of short-answer and extended-response questions evaluating intertwined skills and concepts.
 - **Duration:** 1 hour 30 minutes for SL (50 marks) | 2 hours 30 minutes for HL (90 marks).

Internal Assessment (20% of final grade)

- **The Scientific Investigation:** An individual research report (maximum 3,000 words) evaluating a student-designed experiment or inquiry.
- **Structure & Criteria:** Graded out of 24 marks based on four core criteria: *Research Design* (6 marks), *Data Analysis* (6 marks), *Conclusion* (6 marks), and *Evaluation* (6 marks). Students may collaboratively work in small groups to gather lab data, provided each individual focuses on a distinct independent variable and submits a unique analysis.